Geometry-MMALS G1 v1.0.6

Static Route Controls and Counterbalanced Curriculum

Document	Archival protocol report
Scientific status	C0 protocol implementation only
Date	14 June 2026
Primary notebook	Geometry_MMALS_G1_StaticRoutes_CounterbalancedCurriculum_v1_0_6.ipynb

Purpose

v1.0.6 tests whether instance-adaptive routing adds functional value beyond uniform and learned global static host mixtures, and whether curriculum effects persist after the final task and serial positions are controlled.

This document specifies the experiment and archive. It does not claim C1-C6 or adaptive-route superiority.

1. Evidence motivating the revision

v1.0.5 observation	Seed-0 value	Why v1.0.6 changes the protocol
Geometry context-order delta	+0.0103	Small geometric effect; requires utility controls
Anchor interpolation delta	about +0.085	Functional retention matters more than geometry alone
Adaptive standard accuracy	about 0.521	Baseline for router interventions
Router-z0 zeroed accuracy	about 0.617	Context/static-like routing may be better
Both router inputs zeroed	about 0.611	A fixed route prior may outperform adaptation
Context/z0 route-shift ratio	about 0.148	Direct sensory path dominates routing
Best old curriculum interpolation	about 0.704	Terminal task was confounded

2. Route-policy controls

The static controls use the same sensory encoder, hosts, classifier, paired views, anchor loss, image-forward budget, optimizer-step budget, and source order as the adaptive no-geometry anchor control.

Variant	Route policy	Geo	Anchor	Purpose
current_only_step_matched	adaptive z0 + context	0	0	cheaper step-matched reference
paired_compute_matched_no_geometry	adaptive z0 + context	0	0	paired compute control
paired_geometry	adaptive z0 + context	0.10	0	geometry attribution
paired_anchor_adaptive_no_geometry	adaptive z0 + context	0	0.10	adaptive utility control
paired_geometry_anchor_010	adaptive z0 + context	0.10	0.10	geometry plus retention
paired_anchor_context_only	adaptive context only	0	0.10	context sufficiency
paired_anchor_direct_z0	adaptive z0 only	0	0.10	sensory bypass
paired_anchor_uniform_static	fixed equal mixture	0	0.10	simplest static control
paired_anchor_learned_static	one learned global mixture	0	0.10	optimized static control

The learned static route starts from zero logits, which is exactly the uniform simplex point. Any later difference is learned from the shared training objective.

Primary adaptive-route utility contrasts

- **Adaptive versus uniform:** paired_anchor_adaptive_no_geometry - paired_anchor_uniform_static.
- **Adaptive versus learned static:** paired_anchor_adaptive_no_geometry - paired_anchor_learned_static.
- **Geometry under anchor:** paired_geometry_anchor_010 - paired_anchor_adaptive_no_geometry.

3. Same-final-task counterbalanced curriculum

All four curricula end at 30 degrees. The first four positions form a cyclic counterbalance: every other angle appears once in every pre-final position.

Name	Position 1	Position 2	Position 3	Position 4	Final
cb_a	-60	-30	0	60	30
cb_b	-30	0	60	-60	30
cb_c	0	60	-60	-30	30
cb_d	60	-60	-30	0	30

The notebook fails before training if angle coverage, common final task, or position balance is violated.

4. Router intervention evidence

Condition	Router input	Question
adaptive_standard	z0 + context	reference adaptive policy
router_context_zeroed	z0 + 0	is context necessary?
router_context_shuffled	z0 + another source context	is context source-specific?
router_z0_zeroed	0 + context	is context sufficient?
router_both_inputs_zeroed	0 + 0	does a static router prior work better?

Each intervention is first averaged per source across angles. Accuracy delta, class-evidence change, and route shift then receive paired bootstrap intervals across source IDs.

5. Evidence ladder and decision rules

Decision	Required evidence
C0 protocol integrity	version equality, source hashes, finite-value checks, budget checks, static-route checks, curriculum balance, complete exports
C1 geometry	trained-angle source-block and centroid evidence with frozen multi-seed intervals
C2 interpolation	source-paired held-out accuracy and route interpolation gains
Adaptive-route utility	adaptive policy beats both uniform and learned-static controls at equal compute
Curriculum robustness	bounded performance variation across cb_a-cb_d with common final task
C3 specialization	route mass plus functional ablation, locality, and cross-seed role matching
C4 causality	adaptive policies only: CSR, orientation, monotonicity, identity preservation, source-level CI
C5 memory transport	reconstructive and synthetic memory experiments
C6 operational utility	accuracy-retention-cost benefit over hardened controls

Evidence partitions

- **C1 trained:** -60, -30, 0, 30, 60 only.
- **C2 interpolation:** -45, -15, 15, 45 only.
- **Extrapolation:** -75 and 75, reported separately.
- **Static policy rule:** static routes are excluded from causal context gates.
- **Uncertainty rule:** source blocks are the resampling unit; dependent all-pairs p-values are not evidence.

6. Release integrity and archive

Path	Purpose
notebooks/Geometry_MMALS_G1_StaticRoutes_CounterbalancedCurriculum_v1_0_6.ipynb	executable Colab protocol with history and CHG-106 tracking
docs/reports/Geometry_MMALS_G1_v1_0_6_Static_Routes_Counterbalanced_Curriculum_Report.pdf	archival report
docs/reports/...Report.md	editable source
docs/changes/Geometry_MMALS_G1_v1_0_5_to_v1_0_6.md	tracked delta
configs/rotated_mnist_g1_v106.yaml	declared controls and curricula

The package declares version 1.0.6 and excludes repository metadata, datasets, runtime outputs, caches, bytecode, editable-install metadata, and local environments.

7. Tracked changes and qualification boundary

- CHG-106-01 release integrity guard
- CHG-106-02 uniform static route
- CHG-106-03 learned global static route
- CHG-106-04 adaptive no-geometry anchor control
- CHG-106-05 adaptive-route utility contrasts
- CHG-106-06 static initialization audit
- CHG-106-07 clear intervention names
- CHG-106-08 source-paired intervention intervals
- CHG-106-09 common final task
- CHG-106-10 cyclic counterbalanced positions
- CHG-106-11 curriculum balance assertions
- CHG-106-12 separated geometry partitions
- CHG-106-13 adaptive-only causal probe
- CHG-106-14 static host evidence
- CHG-106-15 clean evidence export
- CHG-106-16 strict non-promotion

Qualification boundary

v1.0.6 is ready for smoke and development execution. No adaptive-route, curriculum-robustness, or C1-C6 claim is allowed until the exact release is executed, gates are frozen, the sensory gate passes, five pilot seeds and ten final seeds are aggregated, failed runs are reported, adaptive routing beats both static controls, the counterbalanced curricula produce bounded variation, and a second transformation is replicated.